Urdu BPE Tokenizer — Final Report

Q1 | Member A — Corpus collection, cleaning, and preparation

Corpus: 1,094.72 MB cleaned | 304,040 documents | 908,493 unique words

Member A's Q1 work assembled a multi-source Urdu corpus and produced document-level 98/1/1 train/validation/test splits (seed 42). The final split contains 1,094.72 MB of UTF-8 text.

Source composition

Source	Genre	Cleaned MB	Documents
jang	news	25.90	12,287
independent urdu	news	23.93	3,709
express live	news	0.53	7
ufal urdu mono	mixed (multiple domains)	715.75	109,292
urdu wikipedia	encyclopedia	328.61	178,745

Self-collected by scraping: 51.02 MB raw / 50.36 MB cleaned across three news sources.

Cleaning and split

Extract article text and remove URLs/emails; apply Urdu Unicode normalization; filter lines below 50% Urdu-script letters and documents under 100 characters; exact-deduplicate; remove recurring short boilerplate; split by document and append <|endoftext|>.

Normalization maps Arabic variants to Urdu forms, removes inconsistent harakat/tatweel and invisible controls, maps Arabic-Indic digits to ASCII, and removes ZWNJ. These choices reduce spelling fragmentation but mean raw, marked text is outside the training distribution.

Q1 evidence and limitations

The source register and corpus data card retain unresolved publisher-terms and collection-date fields for some scraped sources. The raw crawl-evidence directories are not present in this workspace. These fields/evidence should be completed and checked before redistribution; this report does not infer missing licence permissions.

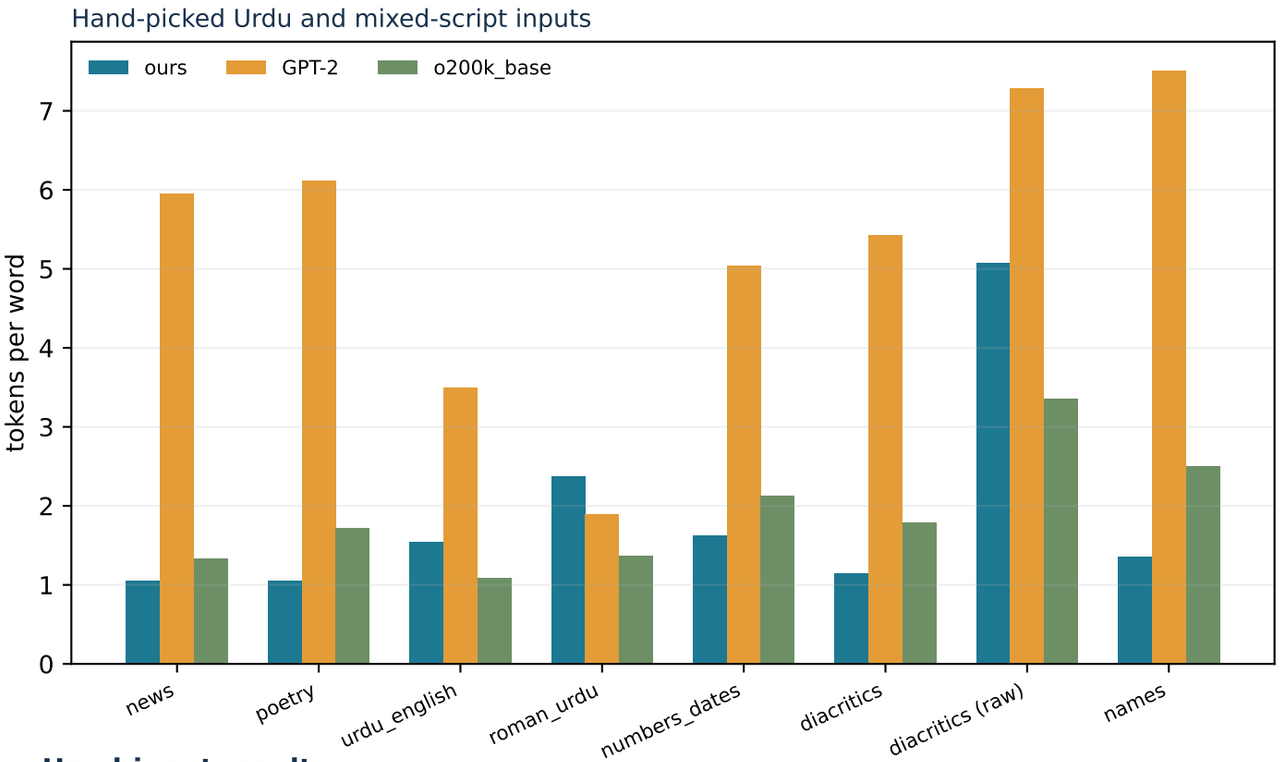
Source: corpus/DATA_CARD.md, corpus/stats.json, and sources.json.

Q2 | Comparison with GPT-2 and o200k_base

All 3,093 documents in corpus/test.txt; whitespace-separated words are the denominator

1,410,766 words | 11,082,910 UTF-8 bytes | 1.17 ours vs 5.692 GPT-2 vs 1.545 o200k_base tokens/word

On this test split, ours uses 24.3% fewer tokens per word than o200k_base and 79.4% fewer than GPT-2. Bytes/token: 6.716 ours, 1.38 GPT-2, 5.085 o200k_base.



Hand-input results

Input	Form	Ours	GPT-2	o200k_base
news	normalised	1.047	5.953	1.326
poetry	normalised	1.056	6.111	1.722
urdu_english	normalised	1.542	3.5	1.083
roman_urdu	normalised	2.368	1.895	1.368
numbers_dates	normalised	1.625	5.042	2.125
diacritics	normalised	1.143	5.429	1.786
diacritics (raw)	raw	5.071	7.286	3.357
names	normalised	1.357	7.5	2.5

Tokenizer comparison uses tiktoken only for GPT-2/o200k_base; our tokenizer uses the trained 50k vocabulary.

Q3 | Three tokenizer failure cases

Failure cases are measured on the same hand-picked inputs from report/hand_inputs.json

1. Roman Urdu (Latin-script Urdu)

Ours needs 45 tokens for 19 whitespace words (2.368 tokens/word), versus 36 for GPT-2 and 26 for o200k_base. Latin spelling variants are poorly represented in a predominantly Urdu-script training corpus.

Mitigation: add licensed Roman Urdu/code-switch data or normalize/transliterate it consistently.

Measured ours/GPT-2/o200k_base: 2.368 / 1.895 / 1.368 tokens per word.

2. Raw Urdu with diacritics

The raw marked sample takes 71 tokens / 14 words (5.071 tokens/word). The normalized form takes 16 / 14 (1.143 tokens/word). This is expected from the preprocessing policy, but raw inputs incur severe fragmentation.

Mitigation: apply the documented normalizer at inference, or retrain with marked-text coverage.

Measured ours/GPT-2/o200k_base: 5.071 / 7.286 / 3.357 tokens per word.

3. Urdu-English code-switching

Ours takes 37 tokens / 24 words (1.542 tokens/word), compared with 26 for o200k_base. English words fragment (for example, laptop becomes l | ap | top and Python becomes P | yth | on), even though ours remains much better than GPT-2 on this sample.

Mitigation: include representative code-switch and English technical vocabulary in training.

Measured ours/GPT-2/o200k_base: 1.542 / 3.5 / 1.083 tokens per word.

Interpretation

These are targeted stress cases, not estimates of prevalence in test.txt. The held-out test average remains 1.170 tokens/word. Results depend on normalization: the raw diacritics row is intentionally scored without normalization while the standard corpus is normalized.